

Confidence intervals. Population known, z-score

Data scientist salary

Background You are given the same dataset from the lesson. The population standard deviation is known to be \$ 15,000.

Task 1 Calculate the mean and the standard error

Task 2 Find the appropriate z-score for calculating a 90% confidence interval

Task 3 Find the 90% confidence interval

Solution:

Dataset

\$ 117,313
\$ 104,002
\$ 113,038
\$ 101,936
\$ 84,560
\$ 113,136
\$ 80,740
\$ 100,536
\$ 105,052
\$ 87,201
\$ 91,986
\$ 94,868
\$ 90,745
\$ 102,848
\$ 85,927
\$ 112,276
\$ 108,637
\$ 96,818
\$ 92,307
\$ 114,564
\$ 109,714
\$ 108,833
\$ 115,295
\$ 89,279
\$ 81,720
\$ 89,344
\$ 114,426
\$ 90,410
\$ 95,118
\$ 113,382

Task 1: Sample mean \$ 100,200
Population std \$ 15,000
Standard error \$ 2,739

Task 2: 90% CI, $z_{0.05}$ 1.65

$$\text{Margin of Error} = z \times SE$$

$$\text{Confidence Interval} = \bar{x} \pm (z \times SE)$$

Task 3:

Z	CI low	CI high
90%	\$ 95,682	\$ 104,719

SE (Standard Error): How much sample average deviates from the population average

Z-score: The confidence number of being sure of the null hypothesis holding i.e. z-score outside of this value means the null hypothesis is unlikely.

The **Margin of Error** is used in calculating the Confidence Interval i.e. the range by which the sample mean can possibly vary by plus or minus and therefore the range where the answer lies.

For example, if a poll says 55% of people like Coke with a Margin of Error of 3%, it means the real number is likely between 52% and 58%.